

Dr. Ebuka Philip Oguchi Ph.D.

CONTACT INFORMATION

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ACADEMIC APPOINTMENT

Assistant Professor (Tenure Track)
Department of Computer and Information Sciences
Spelman College
January 2026 – Present

RESEARCH INTERESTS

Cybersecurity; authentication and message integrity verification; trust establishment for emerging wireless systems; physical-layer security; wireless and mobile security; Agricultural Internet of Things; autonomous and vehicular network security; trustworthy AI; distributed confidential computing; RF fingerprinting; location verification; molecular and nano-scale communication security; applied machine learning for adversarial environments; cyber-physical systems.

EDUCATION

University of Nebraska–Lincoln
Ph.D. in Computer Science, August 2025
Advisor: Prof. Nirnimesh Ghose
Dissertation:
Authentication and Message Integrity Verification for Emerging Wireless Networks
GPA: 4.0/4.0

Changchun University of Science and Technology, China

M.S. in Computer Applied Technology, 2020
Advisor: Prof. Fang Ming
Thesis:
Smoke Recognition Using ResNet and GoogleNet

Nnamdi Azikiwe University, Nigeria

B.Eng. in Electronic and Computer Engineering, 2016
Senior Project:
Biometric Course Registration System Using Fingerprint Authentication

RESEARCH SUPPORT

- **Senior Personnel/ Collaborator**, NSF Center for Distributed Confidential Computing (CDCC) 2026–Present
Multi-institutional NSF Frontiers Center supporting research in confidential computing, secure distributed systems, and trustworthy computing.

- **Collaborator**, Google.org Cybersecurity Clinic 2026–Present
Supporting the establishment of the Spelman College Cybersecurity Clinic and integration of clinic activities into the undergraduate Cybersecurity curriculum.
- **Graduate Research Support**
Supported through NSF awards CNS-2331191, CNS-2225161, CNS-1619285, and ECCS-2030272.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
Soil-Assisted Trust Establishment for Underground Wireless Networks.
IEEE Transactions on Dependable and Secure Computing (TDSC).
Accepted for Publication, 2026.
2. Anderson, M.*; Duong, T.T.*; **Oguchi, E.***; Wisniewska, A.; Ghose, N.
A Review of Confidentiality, Integrity, and Availability Challenges in Molecular Communications.
IEEE Transactions on Molecular, Biological, and Multi-Scale Communications.
Accepted for Publication, 2026.

Peer-Reviewed Conference Papers

1. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
VET: Autonomous Vehicular Credential Verification Using Trajectory and Motion Vectors.
EAI SecureComm, Hong Kong, 2023.
2. **Oguchi, E.**; Ghose, N.; Vuran, M.C.
STUN: Secret-Free Trust Establishment for Underground Wireless Networks.
IEEE INFOCOM Workshops, 2022.

Manuscripts Under Review

1. **Oguchi, E.**; Lado, H.; Evans, R.; Menon, A.; Wang, B.; Ghose, N.
LocaRadio: RF Fingerprint-Based Location Authentication for Wireless Networks.
Submitted to IEEE CCNC.
*Equal contribution.

Dissertation and Theses

1. **Oguchi, E.**
Authentication and Message Integrity Verification for Emerging Wireless Networks.
Ph.D. Dissertation, University of Nebraska–Lincoln, 2025.
2. **Oguchi, E.**
Smoke Recognition Using ResNet and GoogleNet.
M.S. Thesis, Changchun University of Science and Technology, 2020.

3. Oguchi, E.

Biometric Course Registration System Using Fingerprint Authentication.

B.Eng. Project, Nnamdi Azikiwe University, 2016.

RESEARCH EXPERIENCE

Research Assistant

2021–2025

Department of Computer Science and Engineering
University of Nebraska–Lincoln

- Developed the **STUN (Secret-Free Trust Establishment for Underground Wireless Networks)** protocol, a physical-layer authentication framework for secure bootstrapping in Agricultural Internet of Things (Ag-IoT) environments. The work was published at IEEE INFOCOM Workshops and later extended into an accepted journal article in IEEE Transactions on Dependable and Secure Computing.
- Developed the **VET (Vehicular Credential Verification)** framework for autonomous and connected vehicle networks by integrating trajectory information, motion vectors, Doppler characteristics, and wireless signal measurements to verify transmitter authenticity.
- Investigated RF fingerprinting and location authentication techniques using wireless channel characteristics, signal processing, and machine learning for underground and over-the-air communication environments.
- Conducted research on lightweight security mechanisms for molecular and nano-scale communication networks, including authentication, message integrity, and threat modeling for biological communication systems.
- Designed and implemented wireless experimental testbeds using Software Defined Radios (USRPs), GNU Radio, MATLAB, Python, and embedded hardware platforms.
- Contributed to externally funded NSF research projects involving secure wireless systems, underground communications, cyber-physical systems, and confidential distributed computing.

TEACHING EXPERIENCE

Assistant Professor (Tenure Track)

Jan. 2026–Present

Department of Computer and Information Sciences
Spelman College

- SCIS 216: Computer Organization and Design
- SCIS 216L: Computer Organization and Design Laboratory
- SCIS 123: C++ Programming
- Developed laboratory exercises involving digital logic, assembly language programming, breadboard implementation, and computer architecture.
- Mentored undergraduate students through course projects, independent studies, and research activities.

Teaching Assistant

Fall 2022

Cryptography and Security (CSCE 477/877)
University of Nebraska–Lincoln

- Assisted with laboratories covering RSA, AES, hash functions, digital signatures, authentication protocols, and network security.

- Guided students in implementing cryptographic algorithms and analyzing protocol security.

Teaching Assistant

Jan. 2018–May 2018

Computer Architecture and Software Laboratory
D.S. Adegbenro ICT Polytechnic, Nigeria

UNDERGRADUATE RESEARCH MENTORING	• Mentor undergraduate students through independent studies and faculty-directed research projects. • Supervise projects involving wireless security, cybersecurity, Software Defined Radios, embedded systems, machine learning, and trustworthy AI. • Mentor students through literature reviews, software implementation, experimental design, data analysis, manuscript preparation, and conference presentations. • Support students pursuing Honors theses and undergraduate research experiences. • Prepare students for graduate school through research advising, fellowship applications, technical writing, and presentation skills. • Encourage participation in undergraduate research symposia, conference poster sessions, and peer-reviewed publications whenever appropriate.
PROFESSIONAL SERVICE	• Reviewer, IEEE Transactions on Dependable and Secure Computing (2026–Present) • Reviewer, IEEE Transactions on Industrial Informatics • Reviewer, ACM Transactions on Cyber-Physical Systems • Reviewer, ACM WiSec • Reviewer, EAI SecureComm • Judge, Spelman College Hack-A-Bot Competition (2026) • Judge, University of Nebraska Engineering Poster Competition • Member, Departmental Curriculum Review Committee, Spelman College
PROFESSIONAL DEVELOPMENT	• Fellow, ACS Teaching and Learning Workshop, Rollins College (2026) • Faculty Onboarding and Mentorship Program, Spelman College (2026–Present) • CITI Responsible Conduct of Research Training
PROFESSIONAL MEMBERSHIPS	• Institute of Electrical and Electronics Engineers (IEEE) • National Society of Black Engineers (NSBE) • Google Computer Science Research Mentorship Program (CSRMP) • Institute for African-American Mentoring in Computing Sciences (iAAMCS)

AWARDS AND
HONORS

- Outstanding Ph.D. Dissertation Award, University of Nebraska–Lincoln (2025)
- Mary E. and Elmer H. Dohrmann Fellowship, University of Nebraska–Lincoln (2023)
- Chinese Government Scholarship (2018)
- Outstanding Graduate Student Recognition, University of Nebraska–Lincoln

INVITED
ACTIVITIES

- Senior Personnel, NSF Center for Distributed Confidential Computing (CDCC)
- Collaborator, Google.org Cybersecurity Clinic Initiative
- Invited Judge, Spelman College Hack-A-Bot Competition
- Poster Judge, University of Nebraska College of Engineering Research Fair

TECHNICAL SKILLS **Programming:** Python, C, C++, MATLAB, Java, Assembly (MIPS)

Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn

Wireless Systems: Software Defined Radio (USRP), GNU Radio, RF Fingerprinting, Wireless Testbeds

Cybersecurity: Cryptography, Authentication, Message Integrity, Network Security, Physical-Layer Security, Penetration Testing, NIST Cybersecurity Framework

Hardware: Embedded Systems, Breadboard Prototyping, Electronic Circuit Design, Microcontrollers

Research Tools: Linux, Git, LaTeX, Microsoft Azure, Docker